

Notes: List (QJE, 2003)

Does Market Experience Eliminate Market Anomalies?

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1 Big picture

- **Question.** Is the endowment effect a stable feature of preferences, or does it shrink as people gain real market experience?
- **Setting.** The paper studies actual behavior in four trading environments: a sportscard show in Orlando, a Disney pin-trading market at Epcot, a one-year follow-up with the same sportscard traders, and an auction setting at a Tucson sportscard show. It also reports supporting laboratory evidence and a simple equilibrium discussion.
- **Core challenge.** Experienced traders may look more neoclassical either because experience changes behavior (*treatment*) or because people who naturally lack the anomaly are the ones who choose to trade more often (*selection*).
- **Main conceptual contribution.** The paper moves the endowment-effect debate from a pure laboratory setting into naturally functioning markets, where people self-select into trading and differ sharply in actual market sophistication.
- **Main hypotheses.**
 - If the endowment effect is stable, then trading experience should be unrelated to the probability of trade or to the size of the WTA/WTP gap.
 - If experience attenuates the anomaly, then more experienced traders should trade closer to the neoclassical benchmark of 50% in the exchange experiments.
 - If selection matters but treatment does not, then the same individuals should not become more willing to trade as they accumulate experience over time.
- **Main empirical design.** Subjects are randomly endowed with one of two similar goods and then offered a costless swap for the alternative good. The paper compares observed trade rates to the 50% no-endowment-effect benchmark, estimates logit models of the trade decision on experience and controls, and then uses follow-up data, a sample-selection correction, and panel fixed-effects methods to address treatment versus selection.
- **Main baseline result.** In the pooled sportscard sample, only 32.8% and 34.6% of subjects trade, and in the pooled pin-market sample only 25.0% and 32.5% trade. Both are far below the 50% benchmark, so the aggregate data show a clear endowment effect.
- **Key heterogeneity result.** The anomaly is concentrated among inexperienced traders. In the first sportscard market, experienced dealers trade at rates around 44–46%, while inexperienced nondealers trade at only 20–26%; among the least experienced nondealers the trade rate falls to 6.8%.
- **Key treatment-versus-selection result.** In the follow-up experiment with the same nondealers one year later, subjects who gained trading experience became more likely to trade; panel models with individual fixed effects still find a positive and significant effect of experience.
- **Key institutional robustness result.** In the auction environment, nondealers display a very large value gap (mean WTA \$18.53 versus mean WTP \$3.32), while dealers show a much smaller and statistically insignificant gap (mean WTA \$8.15 versus mean WTP \$6.27).

- **Economic takeaway.** Market anomalies can be large for naive agents but much smaller for practiced market participants. The paper therefore suggests that at least part of the endowment effect is shaped by market experience rather than by immutable preferences.

2 Data and measurement (key objects)

2.1 Why these natural markets are useful laboratories

- The central design choice is to observe people in places where they already go to trade, rather than in an artificial lab market that temporarily assigns them trader roles.
- The sportscard and Disney pin markets provide large natural variation in trading intensity, years of market experience, and sophistication.
- The market setting also makes the experiment minimally obtrusive: subjects are asked to do something very similar to what they came to the venue to do anyway.
- Because the goods are unique memorabilia items, subjects are experienced in the *market* but not in repeated rounds with the exact same object.

Interpretation. This is why the paper is important methodologically: it studies whether market sophistication learned in the field can undo a well-known laboratory anomaly.

2.2 Experiment I: sportscard market in Orlando

- The first field experiment was conducted at a sportscard show in Orlando, Florida.
- The sample contains 148 subjects: 74 dealers and 74 nondealers.
- Good A is a Cal Ripken, Jr. ticket stub from June 14, 1996, the game in which Ripken broke the consecutive-games-played record.
- Good B is a Nolan Ryan commemorative certificate from July 31, 1990, celebrating his 300th win.
- A pretest with 50 people suggested the two goods were similar in value: 27 hypothetically chose the Ripken item and 23 chose the Ryan item.
- Subjects first completed a short survey and received one good as payment. They physically held the endowed good while completing the survey and were then offered a direct swap for the other good.
- Nondealers were assigned treatment by time slot; dealers were visited at their booths before the show opened, with the endowed good alternated across booths.

Interpretation. The design deliberately makes the initial endowment psychologically salient. If experienced traders still exchange at close to 50% despite this, that is informative evidence against a stable anomaly.

2.3 Experiment II: Disney pin market at Epcot

- The second field experiment was conducted in the Walt Disney World pin-trading market at Epcot in early May 2000.
- The sample contains 80 consumers. The paper does not split this market into dealers and nondealers because there are very few true dealers in the pin market.
- Good C is a Valentine’s Day 2000 cloisonné pin showing Mickey and Minnie Mouse.
- Good D is a St. Patrick’s Day 2000 cloisonné pin showing Mickey Mouse.
- Each pin retailed for roughly \$20 and sold out quickly, which made them plausible candidate goods of comparable value.
- The protocol mirrored Experiment I: endowment, survey, inspection of the alternative good, swap decision, and exit interview.

Interpretation. This market is demographically different from the sportscard market, especially because it includes many more women, so it provides an important robustness check rather than a simple replication.

2.4 Experiment III: one-year follow-up with the same sportscard subjects

- To separate treatment from selection, the author returned roughly one year later to the sportscard market and tried to recruit the original subjects again.
- Of the 148 original participants, 108 agreed to participate and 72 ultimately appeared: 53 nondealers and 19 dealers.
- The core analysis focuses on the 53 nondealers because the learning question is most relevant there.
- Good E is an autographed 5×8 photo of Byron “Mex” Johnson, a Negro League player.
- Good F is an official National League baseball autographed by Byron “Mex” Johnson.
- The follow-up survey added a question asking subjects to explain how and why their trading frequency had changed over the year.

Interpretation. This is the paper’s strongest design element. Because the same people are observed twice, the author can ask whether experience changes behavior within individuals rather than only across individuals.

2.5 Experiment IV: statements of value in a real auction

- The fourth field experiment took place at a sportscard show in Tucson, Arizona, in September 2000.
- The object auctioned was a unique sheet of University of Wyoming basketball trading cards, including an early Theo Ratliff card.

- The author used a random n th-price auction to elicit either willingness to pay (WTP) or willingness to accept (WTA).
- The design generated 120 observations, split evenly across dealer and nondealer groups and across WTP and WTA elicitation: 30 dealer WTA, 30 dealer WTP, 30 nondealer WTA, and 30 nondealer WTP.
- The author informed subjects about the auction rules and emphasized that truth-telling was the optimal strategy.

Interpretation. The auction evidence matters because it asks whether experience reduces the anomaly in an institution based on explicit value statements, not just simple barter exchange.

2.6 Key variables

The paper's main objects are straightforward and consistent across experiments.

- **Trade indicator.** Equals one if the subject swaps the endowed good for the alternative good, and zero otherwise.
- **Trading experience.** Number of trades the subject reports making in a typical month.
- **Years of market experience.** Number of years the subject has been active in the relevant market.
- **Income.** An ordered category from 1 to 8.
- **Age.** Actual age in years.
- **Gender.** Indicator equal to one for male.
- **Education.** Ordered education category.
- **Good indicator.** Dummy for whether the subject received a particular initial endowment (Good B, D, or F depending on the experiment).

The paper also creates experienced versus inexperienced groups for descriptive tables:

- In Experiment I, experienced nondealers are those who trade 6 or more times per month.
- In Experiment II, the paper shows robustness using cutoffs of 7 and 5 monthly trades.
- In Experiment III, experienced consumers are those who trade 7 or more times per month.

In the auction setting the key outcome becomes the stated value itself:

- **WTP.** Maximum bid submitted by a subject not initially endowed with the good.
- **WTA.** Minimum compensation required by a subject initially endowed with the good.

Interpretation. The main explanatory variable is not just dealer status. The paper treats experience as a continuous object, which is crucial for asking whether behavior gradually converges to neoclassical predictions.

2.7 Descriptive facts

- In Experiment I, dealers report 14.82 trades per month on average, while nondealers report 5.66.
- Dealers in Experiment I also have more years of market experience: 10.36 years versus 6.95 years for nondealers.
- Pin-market consumers report 6.98 trades per month on average, while follow-up sportscard nondealers report 6.84.
- The pin market has many more women than the sportscard market: only 48% of pin-market consumers are male, versus 86–93% in the sportscard samples.
- Exit interviews suggest that subjects planned to consume rather than immediately resell the items: over 95% in the first sportscard experiment and over 97% in the pin market said they intended to keep the item for themselves.
- The follow-up nondealers were slightly more active traders than the original nondealer sample, but the paper argues the difference is not statistically large enough to overturn the later panel analysis.

Interpretation. The descriptive data already show why the design is powerful: there is substantial experience variation within and across markets, and the markets are not just populated by professional dealers.

3 Models and empirical specifications (all equations + interpretation)

3.1 Nonparametric benchmark for the endowment effect

The simplest prediction comes directly from the random-exchange design. Let a subject be randomly endowed with either good A or good B and then given the option to swap.

If preferences are reversible and there is no endowment effect, then the trade rate should be about one-half:

$$\Pr(\text{Trade}) \approx 0.5. \quad (1)$$

Equivalently, with random assignment and goods of similar value, the share of subjects preferring B when endowed with A should mirror the share preferring A when endowed with B .

Interpretation. Trading rates below 50% indicate undertrading relative to the neoclassical benchmark. The paper formally evaluates this with Fisher's exact test.

3.2 Baseline logit trade equation

The paper next models the trade decision as

$$\text{trade}_i = g(\alpha + \beta_1 T_i + \beta_2 T_i^2 + \beta_3 T_i^3 + \gamma' W_i), \quad (2)$$

where

$$g(m) = \frac{1}{1 + e^{-m}}. \quad (3)$$

Here:

- $\text{trade}_i = 1$ if subject i trades and 0 otherwise;
- T_i is monthly trading experience;
- W_i contains years of market experience, income, age, gender, education, and the endowment-good dummy.

The paper estimates both linear and nonlinear learning specifications and often separates dealers from nondealers.

Interpretation. A positive coefficient on T_i means more experienced subjects are more willing to reverse the initial endowment. Negative coefficients on T_i^2 or positive coefficients on T_i^3 capture diminishing returns or more flexible convergence patterns.

3.3 Selection-corrected follow-up model

Because only some original subjects return for the follow-up experiment, the author estimates a bivariate probit with sample selection:

$$Y_{1i}^* = \beta_1' V_i + \varepsilon_{1i}, \quad (4)$$

$$Y_{2i}^* = \beta_2' Z_i + \varepsilon_{2i}, \quad (5)$$

$$Y_{1i} = 1 \text{ if } Y_{1i}^* > 0, \quad (6)$$

$$Y_{2i} = 1 \text{ if } Y_{2i}^* > 0, \quad (7)$$

with $(\varepsilon_{1i}, \varepsilon_{2i})$ assumed jointly normal.

The first equation models whether the subject reappears in the follow-up sample. The second models whether the subject trades in the follow-up experiment, observed only for those with $Y_{1i} = 1$.

Interpretation. This step asks whether the apparent effect of experience could simply come from more active traders being the ones who returned. If the coefficient on experience remains large after the correction, pure sample-selection stories become less persuasive.

3.4 Panel fixed-effects specification

To control for time-invariant individual heterogeneity, the paper estimates Chamberlain's fixed-effects logit model:

$$\Pr(\text{trade}_{it} = 1) = \frac{\exp(z_{it})}{1 + \exp(z_{it})}, \quad (8)$$

with

$$z_{it} = \alpha + \beta_1 T_{it} + \beta_2 T_{it}^2 + \beta_3 T_{it}^3 + \mu_i, \quad (9)$$

for individual i in experiment t .

Interpretation. The fixed effect μ_i absorbs stable individual traits such as a persistent willingness to

trade or a persistent tendency to avoid trades. What remains is within-person evidence on whether more experience actually changes behavior.

3.5 Auction comparison

In the auction experiment the central object is not the probability of trade but the valuation gap:

$$\text{Gap}_g = E[\text{WTA}_g] - E[\text{WTP}_g], \quad (10)$$

where g indexes the dealer or nondealer group.

Interpretation. A large positive gap is the auction analogue of the endowment effect. Comparing the gap across sophisticated and unsophisticated groups tests whether the experience pattern survives an institutional change.

3.6 Simple market-relevance model

In the discussion section, the paper embeds the phenomenon in a two-good pure-exchange economy. Consumers of type j have Cobb-Douglas utility

$$U_j = G_1^{\phi_j} G_2^{1-\phi_j}, \quad (11)$$

but a fraction Φ of consumers are inexperienced and display downward inertia in one good after an endowment shock.

Interpretation. This simple framework is not used for structural estimation. Its purpose is to show that undertrading by inexperienced agents can move prices and welfare in first-order ways, even if experienced traders behave neoclassically.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline effect

- There is **no instrument** in this paper. Identification comes from randomized endowment assignment and from observed variation in market experience.
- Because subjects are randomly endowed with one of two similar goods, the baseline endowment-effect test is clean: under no anomaly, about half should trade.
- The goods are carefully chosen to be of roughly similar value, and subjects are allowed to inspect both goods before deciding.

Interpretation. The random assignment makes the raw trade-rate evidence persuasive on whether an endowment effect exists in each market.

4.2 Why equal-value concerns are limited

- The paper pretested the two sportscard goods and found near-indifference in hypothetical choice.
- The goods were unique memorabilia items that subjects had not previously traded, which helps isolate *market* experience rather than repeated handling of an identical object.
- The swap itself involved essentially zero transaction costs inside the experiment.
- Since the endowed object was given as payment and held during the survey, the design if anything biases the experiment toward finding an endowment effect rather than against it.

Interpretation. These features make it harder to argue that the high trade rates of experienced traders are due only to one item systematically dominating the other.

4.3 Treatment versus selection

- The big threat to the paper is that more active traders may simply have always been the type of person who does not exhibit an endowment effect.
- The follow-up experiment attacks this directly by studying the same nondealers one year later.
- The bivariate probit sample-selection model addresses whether only certain kinds of subjects returned.
- The Chamberlain fixed-effects logit model then controls for time-invariant unobserved heterogeneity.

Interpretation. The panel evidence does not make experience randomly assigned, but it materially strengthens the case that learning in the market is part of the story.

4.4 Institutional robustness and external validity

- The sportscard barter experiment is replicated in a very different market, Disney pin trading, with similar qualitative results.
- The same experience gradient appears when values are elicited through an auction rather than through simple exchange.
- The paper also reports lab evidence with ordinary goods such as mugs, candy bars, pens, markers, Coke, pencils, highlighters, and letter openers. Trading rates rise over repeated weekly sessions.
- The discussion connects the results to nonexperimental evidence from housing and financial markets.

Interpretation. The paper's argument is not that the anomaly disappears everywhere, but that the direction of change with experience is strikingly stable across environments.

4.5 What the paper cannot fully prove

- Experience is not randomly assigned, so the cross-sectional results alone are not enough to establish causality.

- Even with panel methods, time-varying unobservables could still be correlated with the accumulation of experience.
- The field settings are collector markets, which may not map one-for-one into every other market.
- The simple equilibrium discussion is illustrative, not quantitatively structural.

Interpretation. The paper is strongest as evidence that market experience *attenuates* the endowment effect, not as proof that experience always eliminates it completely in every environment.

4.6 Relation to the literature

- The paper directly engages Knetsch (1989) and Kahneman, Knetsch, and Thaler (1990), which found strong endowment effects in laboratory exchange settings.
- Relative to that literature, List’s main claim is not that the laboratory result is false; it is that the magnitude of the anomaly depends on market sophistication.
- The broader message is consistent with later evidence in housing and finance showing that more sophisticated agents often display less loss-averse or endowment-like behavior.

Interpretation. The paper narrows the gap between behavioral evidence and neoclassical market reasoning by arguing that the anomaly is partly endogenous to experience.

5 Tables and figures

5.1 Table I: participant characteristics across the four field settings

Read:

- Dealers in the first sportscard market are much more experienced than nondealers: 14.82 versus 5.66 trades per month.
- Dealers also have more years of market experience: 10.36 versus 6.95.
- The Disney pin market has a very different composition, especially by gender, with only 48% male participants.
- Follow-up nondealers report 6.84 trades per month, slightly above the original nondealer mean of 5.66.

Economic message. The paper is not comparing two artificially manufactured types. There is real and substantial heterogeneity in market sophistication, both across and within markets, which makes the experience-gradient tests meaningful.

TABLE I
SELECTED CHARACTERISTICS OF PARTICIPANTS

	Sportscard market I		Pin market	Sportscard market II
	Dealers mean (std. dev.)	Nondealers mean (std. dev.)	Consumers mean (std. dev.)	Nondealers mean (std. dev.)
<i>Trading experience</i>	14.82 (11.0)	5.66 (6.42)	6.98 (13.63)	6.84 (7.98)
<i>Years of market experience</i>	10.36 (6.75)	6.95 (9.37)	5.05 (5.64)	7.13 (9.05)
<i>Income</i>	4.26 (1.92)	4.04 (2.06)	4.06 (2.25)	4.36 (1.82)
<i>Age</i>	34.68 (11.98)	34.70 (14.06)	31.48 (13.68)	34.83 (12.51)
<i>Gender (percent male)</i>	0.93 (0.25)	0.86 (0.34)	0.48 (0.50)	0.89 (0.32)
<i>Education</i>	3.42 (1.42)	3.84 (1.49)	3.10 (1.53)	3.85 (1.50)
<i>Good B</i>	0.527 (0.50)	0.527 (0.50)		—
<i>Good D</i>	—	—	0.50 (0.50)	—
<i>Good F</i>	—	—	—	0.53 (0.50)
<i>N</i>	74	74	80	53

a. *Trading experience* represents the number of trades made in a typical month.
b. *Years of market experience* denotes years that the subject has been active in the market.
c. *Income* denotes categorical variable (1–8): 1) Less than \$10,000, 2) \$10,000 to \$19,999, 3) \$20,000 to \$29,999, 4) \$30,000 to \$39,999, 5) \$40,000 to \$49,999, 6) \$50,000 to \$74,999, 7) \$75,000 to \$99,999, 8) \$100,000 or over.
d. *Age* denotes actual age in years.
e. *Gender* denotes categorical variable: 0 if female, 1 if male.
f. *Education* denotes categorical variable 1) Eighth grade or less, 2) High School, 3) 2-Year College, 4) Other Post-High School, 5) 4-Year College, 6) Graduate School Education.
g. *Good B (D) (F)* denotes the subject's initial endowment, and = 1 if the subject was endowed with *Good B (D) (F)*, 0 otherwise.

5.2 Tables II and IV: baseline sportscard evidence

TABLE II
SUMMARY TRADING STATISTICS FOR EXPERIMENT I: SPORTSCARD SHOW

Variable	Percent traded	<i>p</i> -value for Fisher's exact test
Pooled sample (n = 148)		
Good A for Good B	32.8	<0.001
Good B for Good A	34.6	
Dealers (n = 74)		
Good A for Good B	45.7	0.194
Good B for Good A	43.6	
Nondealers (n = 74)		
Good A for Good B	20.0	<0.001
Good B for Good A	25.6	

a. Good A is a Cal Ripken, Jr. game ticket stub, circa 1996. Good B is a Nolan Ryan certificate, circa 1990.
b. Fisher's exact test has a null hypothesis of no endowment effect.

TABLE III
NONDEALER SUMMARY STATISTICS FOR EXPERIMENT I: SPORTSCARD SHOW

Variable	Percent traded	<i>p</i> -value for Fisher's exact test
Experienced nondealers (n = 30)	46.7	0.32
Inexperienced nondealers (n = 44)	6.80	<0.001

a. Experienced nondealers are those consumers who trade 6 or more times per month (5.66 is the mean level of monthly trades for nondealers). Inexperienced nondealers trade less than 6 times per month.
b. Fisher's exact test has a null hypothesis of no endowment effect.

Read:

- In Table II, the pooled sportscard trade rates are 32.8% and 34.6%, which rejects the no-endowment-effect benchmark very strongly.
- Dealers trade at 45.7% and 43.6%, and the Fisher exact test does *not* reject the null of no endowment effect ($p = 0.194$).

- Nondealers trade at only 20.0% and 25.6%, and the null of no endowment effect is rejected at $p < 0.001$.
- Table IV then shows that among nondealers the coefficient on monthly trading experience is positive and significant in both the linear and nonlinear logit models.
- For dealers, the experience coefficient is small and statistically weak, consistent with the idea that they are already near the neoclassical benchmark.

Economic message. The anomaly is not uniform across people. It is a strong feature of inexperienced consumers and a weak feature of those who already behave like seasoned traders.

5.3 Tables V and VI: pin-market replication

Variable	Percent traded	<i>p</i> -value for Fisher's exact test
Pooled sample (n = 80)		
Good C for Good D	25.0	<0.001
Good D for Good C	32.5	
Inexperienced consumers (<7 trades monthly; n = 60)	25.0	<0.001
Experienced consumers (≥7 trades monthly; n = 20)	40.0	0.26
Inexperienced consumers (<5 trades monthly; n = 50)	18.0	<0.001
Experienced consumers (≥5 trades monthly; n = 30)	46.7	0.30

a. Good C is a cloisonné Valentine's Day pin portraying Mickey and Minnie Mouse, circa 2000. Good D is a cloisonné St Patrick's Day 2000 portraying Mickey Mouse, circa 2000.
b. Experienced consumers are those consumers who trade 7 (or 5) or more times per month (6.55 is the mean level of monthly trades). Inexperienced consumers trade less than 7 (or 5) times per month.
c. Fisher's exact test has a null hypothesis of no endowment effect.

Variable	Pin consumers		
	Logit trade function	Logit trade function	Logit trade function
<i>Constant</i>	-2.44** (0.91)	-2.57** (0.95)	-4.65 (1.37)
<i>Trading experience</i>	0.05** (0.02)	0.08* (0.05)	0.74** (0.24)
<i>(Trading experience)²</i>	—	-0.004 (0.006)	-0.04** (0.02)
<i>(Trading experience)³</i>	—	—	0.007** (0.003)
<i>Years of market experience</i>	0.03 (0.05)	0.03 (0.05)	0.04 (0.05)
<i>Income</i>	-0.11 (0.18)	-0.10 (0.18)	-0.03 (0.19)
<i>Age</i>	0.005 (0.02)	0.006 (0.03)	0.005 (0.03)
<i>Gender</i>	0.90 (0.55)	0.90 (0.55)	0.41 (0.61)
<i>Education</i>	0.20 (0.23)	0.20 (0.23)	0.26 (0.26)
<i>Good D</i>	0.26 (0.55)	0.29 (0.56)	0.84 (0.63)
<i>N</i>	80	80	80

a. Dependent variable equals 1 if subject chose to trade, 0 otherwise. Gender = 1 if male, 0 otherwise; Good D = 1 if subject was endowed with Good D, 0 otherwise.
b. Standard errors are in parentheses beneath coefficient estimates. Parameter estimates in column 2 are logit coefficients.

Read:

- Table V shows that the pooled pin-market sample also exhibits undertrading: 25.0% and 32.5% trade, again well below 50%.
- But the pattern by experience looks much like the sportscard market. With a 7-trades-per-month cutoff, experienced consumers trade at 40.0% while inexperienced consumers trade at 25.0%.
- With a 5-trades-per-month cutoff, experienced consumers trade at 46.7% while inexperienced consumers trade at only 18.0%.
- Table VI shows that the estimated effect of monthly trading experience is positive, and higher-order terms suggest diminishing returns to additional experience.

Economic message. The sportscard result is not a quirk of one venue or one type of collector. A different market with different demographics delivers the same basic convergence pattern.

5.4 Tables VII, VIII, and X: follow-up and panel evidence

TABLE VII NONDEALER DATA SUMMARY FOR EXPERIMENT III: FOLLOW-UP SPORTSCARD SHOW		
Variable	Percent traded	p-value for Fisher's exact test
Pooled sample (n = 53)		
Good E for Good F	40.0	<0.08
Good F for Good E	35.7	
Experienced consumers (n = 21)		
Good E for Good F	45.5	0.99
Good F for Good E	60.0	
Inexperienced consumers (n = 32)		
Good E for Good F	35.7	<0.02
Good F for Good E	22.2	

a. Good E is an autographed 5 × 8 photo of Byron "Mox" Johnson.
b. Good F is an official National League baseball autographed by Byron "Mox" Johnson.
c. Experienced consumers are those consumers who trade 7 or more times per month. 0.84 is the average level of monthly trades. Inexperienced consumers trade less than 7 times per month.
d. Fisher's exact test has a null hypothesis of no reinforcement effect.

TABLE VIII ESTIMATION RESULTS FOR EXPERIMENT III: FOLLOW-UP SPORTSCARD SHOW			
Variable	Sportscard consumers		
	Logit trade function	Probit trade function	Sample-selection bivariate probit trade function
Constant	-2.40 (1.84)	-1.45 (1.06)	-1.26 (0.98)
Trading experience	0.18** (0.08)	0.112** (0.044)	0.106** (0.040)
Years of market experience	-0.09 (0.09)	-0.06 (0.05)	0.02 (0.05)
Income	0.18 (0.29)	0.09 (0.17)	0.07 (0.15)
Age	-0.05 (0.04)	-0.03 (0.03)	-0.02 (0.02)
Gender	-0.34 (1.03)	-0.15 (0.63)	-0.24 (0.55)
Education	0.52 (0.28)	0.30 (0.16)	0.26 (0.14)
Good F	0.29 (0.78)	0.19 (0.47)	0.16 (0.47)
N	53	53	74

a. Dependent variable equals 1 if subject chose to trade, 0 otherwise. Gender = 1 if male, 0 otherwise; Good F = 1 if subject was matched with Good F, 0 otherwise.
b. Standard errors are in parentheses beneath coefficient estimates. Parameter estimates in column 2 are probit coefficients, while estimates in column 3 are probit coefficients corrected for sample selectivity.
c. **(*) Denotes that coefficient estimate is significant at the p < .05 (.10) level.

TABLE X ESTIMATION RESULTS USING PANEL DATA FROM EXPERIMENTS I AND III						
Variable	Logit trade function			Chamberlain trade function		
	(1)	(2)	(3)	(4)	(5)	(6)
Constant	-1.57** (0.34)	-2.01** (0.44)	-2.91** (0.65)	—	—	—
Trading experience	0.11** (0.04)	0.21** (0.07)	0.55** (0.17)	0.23* (0.12)	0.45** (0.20)	1.33** (0.51)
(Trading experience) ²	—	-0.003* (0.002)	-0.03** (0.01)	—	-0.005* (0.003)	-0.07** (0.03)
(Trading experience) ³	—	—	0.004** (0.002)	—	—	0.009** (0.004)
χ^2 ($\mu_1 = 0$)	—	—	—	3.98**	5.29*	8.47**
N	106	106	106	106	106	106

a. Dependent variable equals 1 if subject chose to trade, 0 otherwise.
b. Standard errors are in parentheses beneath coefficient estimates.
c. **(*) Denotes that coefficient estimate is significant at the p < .05 (.10) level.
d. χ^2 ($\mu_1 = 0$) is a simple Hausman test of the Chamberlain fixed effects model. Each test suggests that there are unobserved fixed effects at the p < .10 level, hence the Chamberlain trade estimates are appropriate.

Read:

- Table VII shows that in the follow-up sample experienced consumers trade at 45.5% and 60.0%, while inexperienced consumers trade at 35.7% and 22.2%.
- Table VIII shows that the coefficient on monthly trading experience remains positive and significant in the follow-up logit, in the simple probit, and in the sample-selection-corrected bivariate probit. The corrected coefficient is 0.106 with a standard error of 0.040.
- Table X is even stronger: once individual fixed effects are introduced through Chamberlain's method, trading experience still predicts trading, and the nonlinear specifications suggest diminishing returns.
- The Hausman-type statistics in Table X reject the no-fixed-effects view, which means that unobserved heterogeneity matters; but even after controlling for it, experience still matters.

Economic message. This is the most important set of tables in the paper. They move the evidence beyond “experienced people are different” toward “the same people become more neoclassical as they gain experience.”

5.5 Figure I: pooled convergence by trading experience

Read:

- The pooled data from the three field trading experiments are grouped by prior trading intensity.
- Subjects with 0–1 trades per month trade only about 0.13 of the time.
- The trade rate rises to roughly 0.21 for those with 2–5 trades, to about 0.34 for those with 6–10 trades, and to around 0.50 for those with 11 or more trades.
- The relationship is monotone and visually very sharp.

Economic message. The figure summarizes the paper's central point in one picture: with enough market experience, behavior converges toward the neoclassical 50% benchmark.

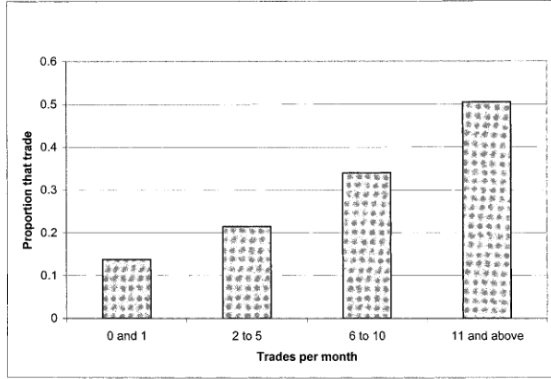


FIGURE I
Summary of Trading Results

TABLE XI
SELECTED CHARACTERISTICS OF TUCSON SPORTSCARD PARTICIPANTS

	Dealers		Nondealers	
	WTA mean (std. dev.)	WTP mean (std. dev.)	WTA mean (std. dev.)	WTP mean (std. dev.)
<i>Bid or offer</i>	8.15 (9.66)	6.27 (6.90)	18.53 (19.96)	3.32 (3.02)
<i>Trading experience</i>	16.67 (19.88)	15.78 (13.71)	4.00 (5.72)	3.73 (3.46)
<i>Years of market experience</i>	10.23 (5.61)	10.57 (8.13)	5.97 (5.87)	5.60 (6.70)
<i>Income</i>	3.46 (2.17)	3.40 (2.03)	3.37 (2.14)	3.40 (2.24)
<i>Age</i>	29.20 (12.20)	31.00 (14.70)	28.40 (14.90)	29.00 (15.30)
<i>Gender (percent male)</i>	0.87 (0.35)	0.90 (0.31)	0.90 (0.31)	0.90 (0.31)
<i>Education</i>	3.36 (1.77)	3.40 (2.03)	3.03 (1.73)	3.23 (1.81)
<i>N</i>	30	30	30	30

a. *Trading experience* represents the number of trades made in a typical month.
b. *Years of market experience* denotes years that the subject has been active in the market.
c. *Income* denotes categorical variable (1–8): 1) Less than \$10,000, 2) \$10,000 to \$19,999, 3) \$20,000 to \$29,999, 4) \$30,000 to \$39,999, 5) \$40,000 to \$49,999, 6) \$50,000 to \$74,999, 7) \$75,000 to \$99,999, 8) \$100,000 or over.
d. *Age* denotes actual age in years.
e. *Gender* denotes categorical variable: 0 if female, 1 if male.
f. *Education* denotes categorical variable 1) Eighth grade or less, 2) High School, 3) 2-Year College, 4) Other Post-High School, 5) 4-Year College, 6) Graduate School Education.

5.6 Table XI: auction WTA versus WTP for dealers and nondealers

Read:

- In the auction setting, dealers' mean WTA is \$8.15 and mean WTP is \$6.27.
- That dealer difference is not statistically significant at conventional levels.
- For nondealers, mean WTA is \$18.53 while mean WTP is only \$3.32.
- The nondealer gap is economically huge and statistically strong.

Economic message. Experience does not just affect simple swapping behavior. It also shrinks the explicit valuation gap that behavioral economics associates with the endowment effect.

6 Short summary

- The paper asks whether the endowment effect survives exposure to real markets.
- Using field experiments in sportscard and Disney pin markets, it first shows that the pooled data do exhibit a standard endowment effect.
- The key result is heterogeneity by experience: dealers and other experienced traders exchange at rates close to the 50% neoclassical benchmark, while inexperienced consumers strongly undertrade.
- The auction experiment reaches the same conclusion using explicit WTA and WTP statements rather than simple barter.
- The follow-up panel with the same subjects is crucial because it shows that growing experience predicts growing willingness to trade even after controlling for fixed individual traits.

- The broader implication is that some market anomalies are not immutable primitives; they can weaken substantially as agents gain practical market experience.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the endowment effect is strong in the aggregate but highly uneven across traders. Inexperienced consumers show large reluctance to trade and large WTA/WTP gaps, while experienced traders behave much more like the neoclassical benchmark predicts. The follow-up panel evidence indicates that this is not only a cross-sectional selection story.

7.2 Economic intuition

Market experience seems to teach people how to view goods less as attached endowments and more as tradable objects. Repeated exposure to exchange, valuation, and bargaining makes traders more willing to reverse an arbitrary initial allocation. Once that happens, trade frequencies move toward 50% and valuation gaps compress.

7.3 Takeaway

List's lasting message is that one should be careful before treating behavioral anomalies as fixed primitives in market settings. Markets populated by naive agents may look very different from markets populated by practiced agents, because experience itself can make behavior more nearly neoclassical.